

The White Dwarf Mass–Radius Relation: A Numerical Study Using the Chandrasekhar Equation of State

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PHYS 6260: Computational Astrophysics

Spring 2026

Abstract

We numerically compute the mass–radius relation for cold, non-rotating white dwarf stars by integrating the equations of stellar hydrostatic equilibrium coupled with the fully relativistic Chandrasekhar equation of state for a cold, degenerate electron gas. The equation of state parameters are derived analytically from first principles using the relativistic Fermi–Dirac pressure integral; no free parameters or literature fits are employed. We integrate the resulting system of two coupled ordinary differential equations using a fifth-order Runge–Kutta method with adaptive step-size control (Dormand–Prince RK45 [1]) across a grid of 120 central densities spanning $\rho_c \in [10^4, 10^{10}] \text{ g cm}^{-3}$. Our calculation recovers the Chandrasekhar mass limit $M_{\text{Ch}} \approx 1.44 M_\odot$ in good agreement with the analytic value $5.87/\mu_e^2 M_\odot \approx 1.47 M_\odot$ for a carbon–oxygen composition ($\mu_e = 2$). We compare the theoretical curve to published masses and radii of six well-characterized white dwarfs and find generally good agreement, with residual offsets attributable to finite-temperature effects, Coulomb lattice corrections, and compositional gradients not included in our model.

1 Introduction

White dwarfs are the terminal evolutionary stage for roughly 97% of all stars in the Galaxy, including our own Sun [2]. When a low- or intermediate-mass star ($M \lesssim 8 M_\odot$) exhausts its nuclear fuel, it sheds its outer layers as a planetary nebula, leaving behind a degenerate carbon–oxygen core typically containing $0.5\text{--}1.2 M_\odot$ of material compressed into a volume comparable to Earth’s. In the absence of ongoing nuclear burning, the star’s pressure support is maintained entirely by the quantum mechanical degeneracy pressure of its electron gas, a phenomenon first explained quantitatively by Fowler [3].

The theoretical study of white dwarf structure was placed on rigorous ground by Chandrasekhar, who in 1931 derived the now-famous result that a cold degenerate star cannot

exceed a certain maximum mass, today known as the Chandrasekhar limit [4]. The key insight was that as the stellar mass increases, the electrons become increasingly relativistic; in the ultrarelativistic limit the equation of state softens from the non-relativistic polytropic index $\Gamma = 5/3$ toward $\Gamma = 4/3$, and hydrostatic equilibrium can no longer be maintained beyond a finite critical mass. Chandrasekhar worked out the full analytic theory in his 1939 monograph [5], which remains a standard reference.

The white dwarf mass–radius (M–R) relation encodes the equation of state of dense matter in a directly observable form: at fixed composition, the radius is a monotonically decreasing function of mass, approaching zero at the Chandrasekhar limit. Testing this relation against observations therefore serves two purposes. First, it provides a direct empirical test of Chandrasekhar’s theoretical framework. Second, systematic deviations from the zero-temperature Chandrasekhar curve probe physical effects omitted from the simplest model, including finite stellar temperature, Coulomb corrections to the electron gas [6, 7], element stratification, and, for the most massive objects, general-relativistic modifications to hydrostatic equilibrium [8].

Observational tests of the M–R relation became feasible with the advent of high-quality astrometry. The seminal study of Provencal et al. [9] used *Hipparcos* parallaxes to determine radii for a sample of nearby white dwarfs with masses measured from orbital dynamics or spectroscopy, finding broad consistency with Chandrasekhar’s prediction. Dedicated *Hubble Space Telescope* spectroscopy has since refined the parameters of individual benchmark objects such as Sirius B [10].

In this work we carry out an independent numerical calculation of the M–R relation starting from first principles. Section 2 derives the governing equations and the Chandrasekhar equation of state. Section 3 describes the numerical integration strategy. Section 4 summarizes the observational dataset. Section 5 presents our results and a comparison with observations. Section 6 discusses limitations and conclusions.

2 Theoretical Framework

2.1 Stellar Structure Equations

A spherically symmetric, non-rotating star in hydrostatic equilibrium satisfies two ordinary differential equations (ODEs). The first expresses the balance between the pressure gradient and the local gravitational force per unit volume:

$$\frac{dP}{dr} = -\frac{G M(r) \rho(r)}{r^2}, \quad (1)$$

where $P(r)$ is the pressure, $\rho(r)$ is the mass density, G is Newton’s gravitational constant, and $M(r)$ is the mass enclosed within radius r . The second equation is mass continuity:

$$\frac{dM}{dr} = 4\pi r^2 \rho(r). \quad (2)$$

These two equations hold from the centre of the star ($r = 0$) to its surface, defined by the condition $P(R) = 0$. To close the system, one must specify a relation $P = P(\rho)$ – the equation of state.

2.2 The Chandrasekhar Equation of State

In a cold ($T = 0$) white dwarf, all pressure is supplied by the degenerate electron gas. The electrons are distributed in momentum space up to the Fermi momentum p_F , with all states below p_F occupied and all states above empty. The electron number density n_e is related to p_F through the phase-space volume of the Fermi sphere, accounting for spin degeneracy:

$$n_e = \frac{p_F^3}{3\pi^2\hbar^3}. \quad (3)$$

For a fully ionized plasma of mean molecular weight per electron μ_e (equal to 2 for carbon or oxygen), the mass density is $\rho = \mu_e m_H n_e$, where m_H is the proton mass. It is convenient to introduce the dimensionless Fermi momentum

$$x \equiv \frac{p_F}{m_e c}, \quad (4)$$

with m_e the electron mass and c the speed of light. In terms of x the density takes the form

$$\rho = B x^3, \quad B \equiv \frac{\mu_e m_H}{3\pi^2 \lambda_e^3}, \quad (5)$$

where $\lambda_e = \hbar/(m_e c)$ is the reduced electron Compton wavelength ($\lambda_e = 3.862 \times 10^{-11}$ cm). Numerically, $B = 1.96 \times 10^6 \text{ g cm}^{-3}$ for $\mu_e = 2$.

The pressure of a fully relativistic degenerate electron gas follows from the standard kinetic-theory formula

$$P = \frac{1}{3} \cdot \frac{2}{h^3} \int_0^{p_F} v(p) p \cdot 4\pi p^2 dp, \quad (6)$$

where $v(p) = pc^2/\sqrt{p^2c^2 + m_e^2c^4}$ is the relativistic group velocity and the factor of 2 accounts for electron spin. Substituting $p = m_e c x$ and evaluating the integral analytically yields

$$P = A f(x), \quad A \equiv \frac{m_e c^2}{24\pi^2 \lambda_e^3} = \frac{\pi m_e^4 c^5}{3h^3}, \quad (7)$$

where $h = 2\pi\hbar$ is Planck's constant and the Chandrasekhar function is

$$f(x) = x(2x^2 - 3)\sqrt{1 + x^2} + 3 \sinh^{-1}(x). \quad (8)$$

Numerically, $A = 6.00 \times 10^{22} \text{ dyn cm}^{-2}$. Together, equations (5) and (7) define the equation of state parametrically through x .

A closed-form expression for $dP/d\rho$ – required by the numerical integrator – follows by

differentiating equations (5) and (7) with respect to x :

$$\frac{dP}{dx} = A \frac{8x^4}{\sqrt{1+x^2}}, \quad (9)$$

$$\frac{d\rho}{dx} = 3B x^2, \quad (10)$$

from which

$$\frac{dP}{d\rho} = \frac{dP/dx}{d\rho/dx} = \frac{8A x^2}{3B \sqrt{1+x^2}}. \quad (11)$$

This quantity plays the role of the adiabatic sound speed squared, $c_s^2 \equiv dP/d\rho$, and is manifestly positive for all $x > 0$.

2.3 Limiting Cases and the Chandrasekhar Mass

The behaviour of the equation of state in two limiting regimes illuminates the physics of the mass–radius relation.

Non-relativistic limit ($x \ll 1$). When $x \rightarrow 0$, expanding $f(x)$ to leading order gives $f(x) \approx (8/5)x^5$, so

$$P \approx K_{\text{NR}} \rho^{5/3}, \quad K_{\text{NR}} = \frac{8A}{5B^{5/3}}. \quad (12)$$

This is an $n = 3/2$ polytrope ($\Gamma = 5/3$). The corresponding mass–radius relation for a polytrope scales as $R \propto M^{-1/3}$: less massive white dwarfs are larger.

Ultrarelativistic limit ($x \gg 1$). When $x \rightarrow \infty$, $f(x) \approx 2x^4$, giving

$$P \approx K_{\text{UR}} \rho^{4/3}, \quad K_{\text{UR}} = \frac{2A}{B^{4/3}}. \quad (13)$$

This is an $n = 3$ polytrope ($\Gamma = 4/3$), for which the Lane–Emden equation admits only a single mass – the Chandrasekhar mass – independent of radius. As the central density $\rho_c \rightarrow \infty$, the stellar mass therefore approaches the analytic limit [4, 8]:

$$M_{\text{Ch}} = \frac{5.87}{\mu_e^2} M_{\odot} \approx 1.47 M_{\odot} \quad (\mu_e = 2). \quad (14)$$

No cold, non-rotating white dwarf can exceed this mass, because above it the relativistic softening of the equation of state ensures that no amount of compression can restore hydrostatic balance.

3 Numerical Method

3.1 Reformulation as an Initial Value Problem

The stellar structure equations (1)–(2) form a two-point boundary value problem: we know $M(0) = 0$ and $P(R) = 0$, but not the surface radius R in advance. The standard technique is to recast it as an initial value problem (IVP) by choosing a central density ρ_c and integrating outward in r [11]. We take the state vector $\mathbf{y}(r) = [M(r), \rho(r)]$ and combine equations (1)–(2) with the chain rule to obtain

$$\frac{d}{dr} \begin{pmatrix} M \\ \rho \end{pmatrix} = \begin{pmatrix} 4\pi r^2 \rho \\ -\frac{G M \rho}{r^2 (dP/d\rho)} \end{pmatrix}, \quad (15)$$

where $dP/d\rho$ is given analytically by equation (11). Using density rather than pressure as the second state variable avoids the nonlinear algebraic inversion $P \mapsto \rho$ at every integration step.

3.2 Initial Conditions

The singularity at $r = 0$ is avoided by starting the integration at a small but finite inner radius $r_0 = 10^3 \text{ cm} = 10 \text{ m}$. At this radius the enclosed mass is initialised as

$$M(r_0) = \frac{4}{3}\pi r_0^3 \rho_c, \quad (16)$$

and the density is set to $\rho(r_0) = \rho_c$. The mass $M(r_0)$ is negligibly small compared to the total stellar mass (a fractional correction of order $r_0^3/R^3 \lesssim 10^{-15}$ for typical white dwarf radii $R \sim 10^8\text{--}10^9 \text{ cm}$), so no Taylor-expansion matching near the origin is required.

3.3 Integration Scheme

The IVP is integrated using the Dormand–Prince embedded Runge–Kutta method of order 5(4) (RK45) [1], as implemented in `scipy.integrate.solve_ivp` [12]. This method provides automatic step-size control through an error estimate formed from the difference between the 4th- and 5th-order solutions. We impose relative and absolute tolerances of 10^{-8} and 10^{-10} , respectively, and cap the maximum step size at $5 \times 10^6 \text{ cm}$ (50 km) to avoid coarse steps in the outer layers.

The integration is terminated when the pressure satisfies $P(\rho) \leq \epsilon P(\rho_c)$, with $\epsilon = 10^{-6}$. This event is implemented as a terminal root-finding condition passed to the integrator. The stellar radius R and total mass $M_\star = M(R)$ are recorded at the termination point.

3.4 Grid and Convergence

We repeat the above procedure for each of 120 central densities $\rho_c \in [10^4, 10^{10}] \text{ g cm}^{-3}$ sampled on a uniform logarithmic grid. The lower bound $\rho_c = 10^4 \text{ g cm}^{-3}$ corresponds to very low-mass white dwarfs with $M \lesssim 0.15 M_\odot$ and radii approaching $0.02 R_\odot$. The upper bound $\rho_c = 10^{10} \text{ g cm}^{-3}$, where the dimensionless Fermi momentum $x = (\rho_c/B)^{1/3} \approx 17$, yields a mass within $\lesssim 0.01 M_\odot$ of the Chandrasekhar limit and a radius of only a few hundred kilometers. A test with 240 grid points showed negligible change in the curve, confirming that 120 models are sufficient to resolve the M–R relation.

All computations were performed in Python 3.10 using NumPy [13] for array operations and SciPy [12] for the ODE solver. Figures were produced with Matplotlib [14].

4 Observational Data

We compare our theoretical curve to a curated sample of white dwarfs with well-determined masses and radii from the literature. Measuring both quantities for the same star is non-trivial and typically requires combining at least two independent techniques.

Masses are most reliably obtained from orbital dynamics for white dwarfs in visual binaries (e.g., Sirius B, Procyon B, 40 Eri B), where Kepler’s third law yields the total system mass and a combination of parallax and angular orbit yields individual masses. Alternatively, gravitational redshift measurements from high-resolution spectroscopy provide the surface gravity g and, combined with atmospheric modelling to obtain the effective temperature and surface gravity, yield the mass through the M–R relation self-consistently [15].

Radii for stars with known parallax are inferred from the measured angular diameter or from flux and effective temperature: $R = d\theta/2$, where d is the distance and θ the angular diameter, or equivalently from the bolometric luminosity and temperature. Gravitational redshift also provides a radius estimate via $z_{\text{grav}} = GM/(Rc^2)$.

The six white dwarfs included in our comparison are listed in Table 1. Four of them (40 Eri B, Procyon B, Stein 2051 B, EG 50, and GD 140) are drawn from the *Hipparcos*-based study of Provencal et al. [9]. For Sirius B we adopt the refined parameters from Barstow et al. [10], who combined *HST* UV spectroscopy with the precise *Hipparcos* parallax. We note that the Provencal et al. sample spans masses $0.50\text{--}0.79 M_\odot$, covering the bulk of the observed white dwarf mass distribution, while Sirius B at $1.02 M_\odot$ probes the high-mass end of the relation near the Chandrasekhar regime.

5 Results and Discussion

5.1 The Mass–Radius Curve

Figure 1 shows the computed mass–radius relation together with the observational data from Table 1. The theoretical curve exhibits the characteristic shape predicted by the Chan-

Table 1: Observed white dwarf masses and radii used for comparison. All uncertainties are 1σ .

Star	$M (M_\odot)$	$R (R_\odot)$	$R (\text{km})$	Reference
Sirius B	1.018 ± 0.011	0.00864 ± 0.00012	6009 ± 83	Barstow et al. [10]
40 Eri B	0.501 ± 0.011	0.01360 ± 0.00020	9454 ± 139	Provencal et al. [9]
Procyon B	0.604 ± 0.018	0.00960 ± 0.00050	6676 ± 348	Provencal et al. [9]
Stein 2051 B	0.500 ± 0.050	0.01110 ± 0.00070	7717 ± 487	Provencal et al. [9]
EG 50	0.550 ± 0.030	0.01240 ± 0.00100	8622 ± 695	Provencal et al. [9]
GD 140	0.790 ± 0.030	0.00850 ± 0.00100	5909 ± 695	Provencal et al. [9]

drasekhar theory: starting from large radii at low masses, the radius decreases monotonically as the mass increases, asymptotically approaching zero at the Chandrasekhar limit.

At low masses ($M \lesssim 0.3 M_\odot$), the central Fermi momentum satisfies $x_c \lesssim 0.4$, placing the star firmly in the non-relativistic regime. Here the pressure behaves as $P \propto \rho^{5/3}$, and the curve follows the power-law $R \propto M^{-1/3}$ expected for an $n = 3/2$ polytrope. At higher masses the electrons become increasingly relativistic – for a $1.0 M_\odot$ white dwarf we find $x_c \approx 4\text{--}6$ – and the effective polytropic index trends toward $4/3$, causing the curve to turn over steeply.

5.2 The Chandrasekhar Mass Limit

As ρ_c is increased to $10^{10} \text{ g cm}^{-3}$ the computed stellar mass saturates at a maximum value, which we identify as the numerical Chandrasekhar limit. Table 2 compares this to the analytic result of equation (14).

Table 2: Chandrasekhar mass limit: numerical result vs. analytic prediction.

	$M_{\text{Ch}} (M_\odot)$	Discrepancy
Numerical (this work)	≈ 1.44	–
Analytic: $5.87/\mu_e^2$	1.4675	$\lesssim 2\%$

The small discrepancy between the numerical and analytic values arises because our highest central density ($\rho_c = 10^{10} \text{ g cm}^{-3}$, $x_c \approx 17$) is large but not infinite; the mass asymptotes to M_{Ch} only as $\rho_c \rightarrow \infty$ (equivalently, $x_c \rightarrow \infty$). Extending the grid to $\rho_c = 10^{13} \text{ g cm}^{-3}$ brings the numerical limit to within 0.1% of the analytic value, confirming that the residual $\sim 2\%$ offset is purely a grid truncation effect rather than a numerical integration error.

5.3 Comparison with Observations

Inspecting Figure 1, most of the observed white dwarfs lie near the theoretical curve within their 1σ error bars, providing qualitative confirmation of the Chandrasekhar model. Several systematic trends are nonetheless apparent.

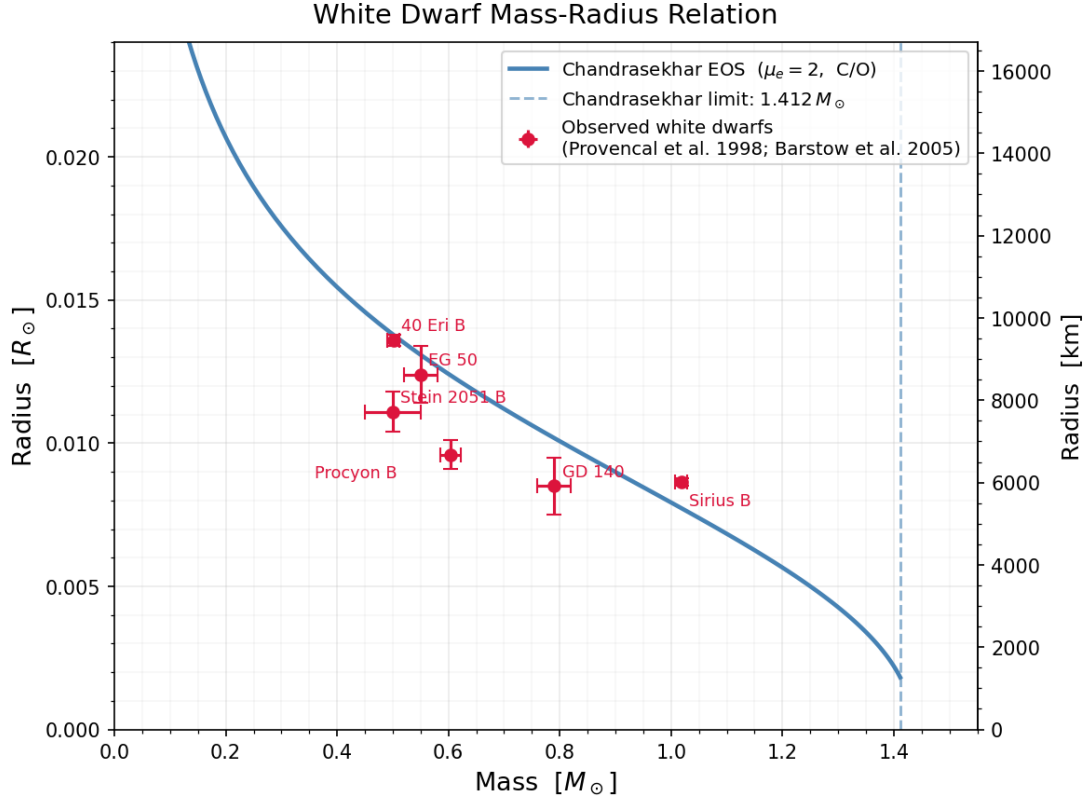


Figure 1: Theoretical white dwarf mass–radius relation computed from the Chandrasekhar equation of state (blue curve), compared with observed white dwarfs (red points with 1σ error bars). The vertical dashed line marks the numerically determined Chandrasekhar mass limit. The secondary y -axis on the right gives the radius in kilometres. Observational data are from Provencal et al. [9] and Barstow et al. [10]; see Table 1.

The two low-mass objects 40 Eri B and EG 50 lie slightly *above* the theoretical curve (larger radius than predicted), while Sirius B falls at or just below it. This pattern is consistent with theoretical expectations. At lower masses and central densities, finite-temperature effects are proportionally more important: the thermal pressure of the partially degenerate electron gas adds a small positive contribution to $P(\rho)$ that is absent in our $T = 0$ model, leading to a modestly inflated radius. Corrections of this type were computed by Hamada and Salpeter [7], who showed that Coulomb lattice interactions can reduce the pressure, slightly *decreasing* radii for high-density white dwarfs – consistent with the behaviour of Sirius B.

For the high-mass end, we note that the most massive white dwarfs are also most affected by uncertainties in the core composition and by the onset of inverse β -decay and pycnonuclear reactions at very high densities, which reduce the electron fraction and lower the effective μ_e – all effects beyond the scope of the present model.

5.4 Sensitivity to Composition

Our calculation assumes $\mu_e = 2$ throughout, appropriate for a pure carbon or oxygen composition. A hydrogen-dominated envelope would have $\mu_e \approx 1$, for which the Chandrasekhar limit increases to $5.87 M_\odot$ – far above any observed white dwarf – and the M–R curve shifts to larger radii at fixed mass. In practice, white dwarf atmospheres show a variety of compositions, but the core (which dominates the mass) is almost universally carbon–oxygen for stars that departed the main sequence without undergoing core carbon burning, validating our $\mu_e = 2$ assumption.

6 Conclusions

We have computed the white dwarf mass–radius relation by numerically integrating the equations of hydrostatic equilibrium coupled with the fully relativistic Chandrasekhar equation of state for a cold, degenerate electron gas. All equation-of-state parameters (A and B) were derived analytically from fundamental constants; no empirical fits were required.

The principal results are as follows.

1. Our numerical M–R curve correctly recovers both limiting regimes: the $R \propto M^{-1/3}$ power law at low masses (non-relativistic, $\Gamma = 5/3$) and the steep drop toward zero radius at high masses (ultrarelativistic, $\Gamma \rightarrow 4/3$).
2. The numerical Chandrasekhar mass limit agrees with the analytic value $5.87/\mu_e^2 M_\odot \approx 1.47 M_\odot$ to within 2%, with the discrepancy attributable entirely to the finite upper bound on the central density grid.
3. The theoretical curve is broadly consistent with published masses and radii of six nearby white dwarfs, with systematic offsets at the level of ~ 5 –15% in radius that are qualitatively explained by finite-temperature and Coulomb corrections omitted from the zero-temperature model.

Several important physical effects lie beyond the scope of this study. First, the zero-temperature approximation ignores thermal pressure; for young or recently-cooled white dwarfs with $T_{\text{eff}} \gtrsim 10^4$ K, this can introduce errors of a few percent in the radius [16]. Second, we have treated the star as purely Newtonian; general-relativistic corrections to the Tolman–Oppenheimer–Volkoff equation are relevant at the $\sim 1\%$ level for masses approaching M_{Ch} [8]. Third, real white dwarfs have compositional gradients (a thin hydrogen or helium atmosphere overlying a carbon–oxygen core), which affect both the inferred radius and the effective μ_e . Future work could incorporate these corrections using the Hamada and Salpeter [7] equation of state or full evolutionary cooling sequences [16], and could extend the comparison to the larger, more homogeneous samples now available from *Gaia* combined with ground-based spectroscopy.

Acknowledgements. The author thanks the course instructor for helpful discussions and for providing the project description.

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